

# Mixed effects models

# Readings for today

- Bates, Douglas M. "lme4: Mixed-effects modeling with R." (2010): 470-474.
- Yarkoni, T. (2019). The generalizability crisis. PsyArXiv

# Topics

1. Fixed vs. random effects
2. Towards increased generalizability
3. Random effects vs. nuisance terms



# Fixed vs. random effects

# Example: random factors

Q: Does having distracting stimuli on classroom walls impact learning?

## Variables:

- $Y_{recall}$ : item recall after 5 minutes
- $X_{walls}$ :  $\begin{cases} 1, & \text{distracting walls} \\ 0, & \text{blank walls} \end{cases}$
- $Z_{class}$ : categorical classroom ID: {336A, 336B, 340A, 329G}

## Bad model:

$$Y_{recall} = \hat{\beta}_0 + \hat{\beta}_1 X_{walls} + \hat{\beta}_2 Z_{class}$$

↑
↑  
 main effect      main effect  
 of wall          of classroom

If classroom ID is an arbitrary label, what does a unit change in  $Z_{class}$  mean with regards to unit changes in  $Y_{recall}$ ?

# Mixed effects models

$$Y = \sum_{j=1}^p \hat{\beta}_j X + \sum_{k=1}^q \hat{v}_k Z$$

*fixed effects*      *random effects*

Fixed Effects: Variables whose relationship with  $Y$  covary in a meaningful way.

*Note: This is how we been thinking about regression variables all along.*

Random Effects: Categorical variables that are not of primary interest but have a systematic influence on  $Y$ .

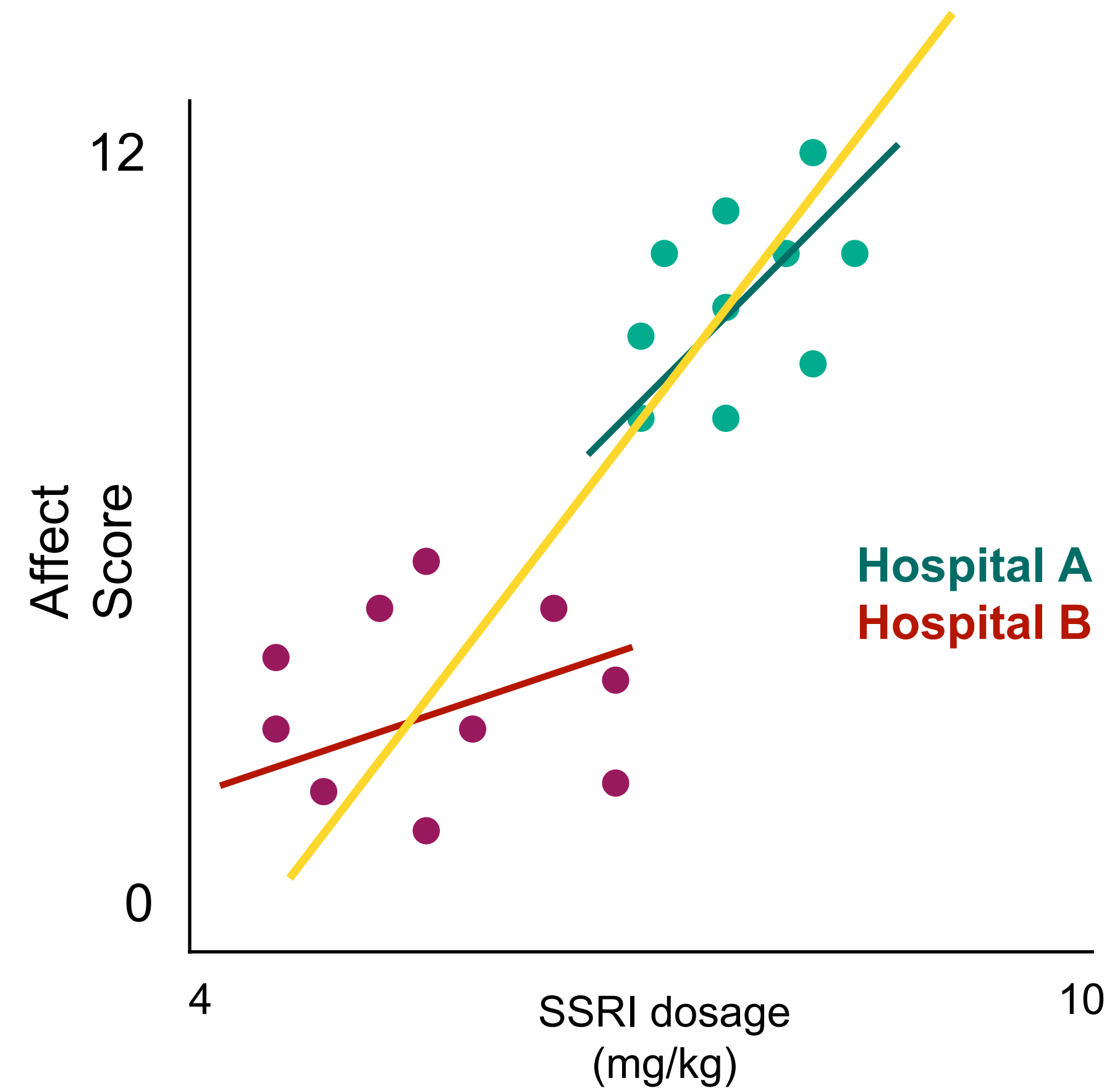


# Example: random effect

Q: Does SSRI dosage increase positive affect?

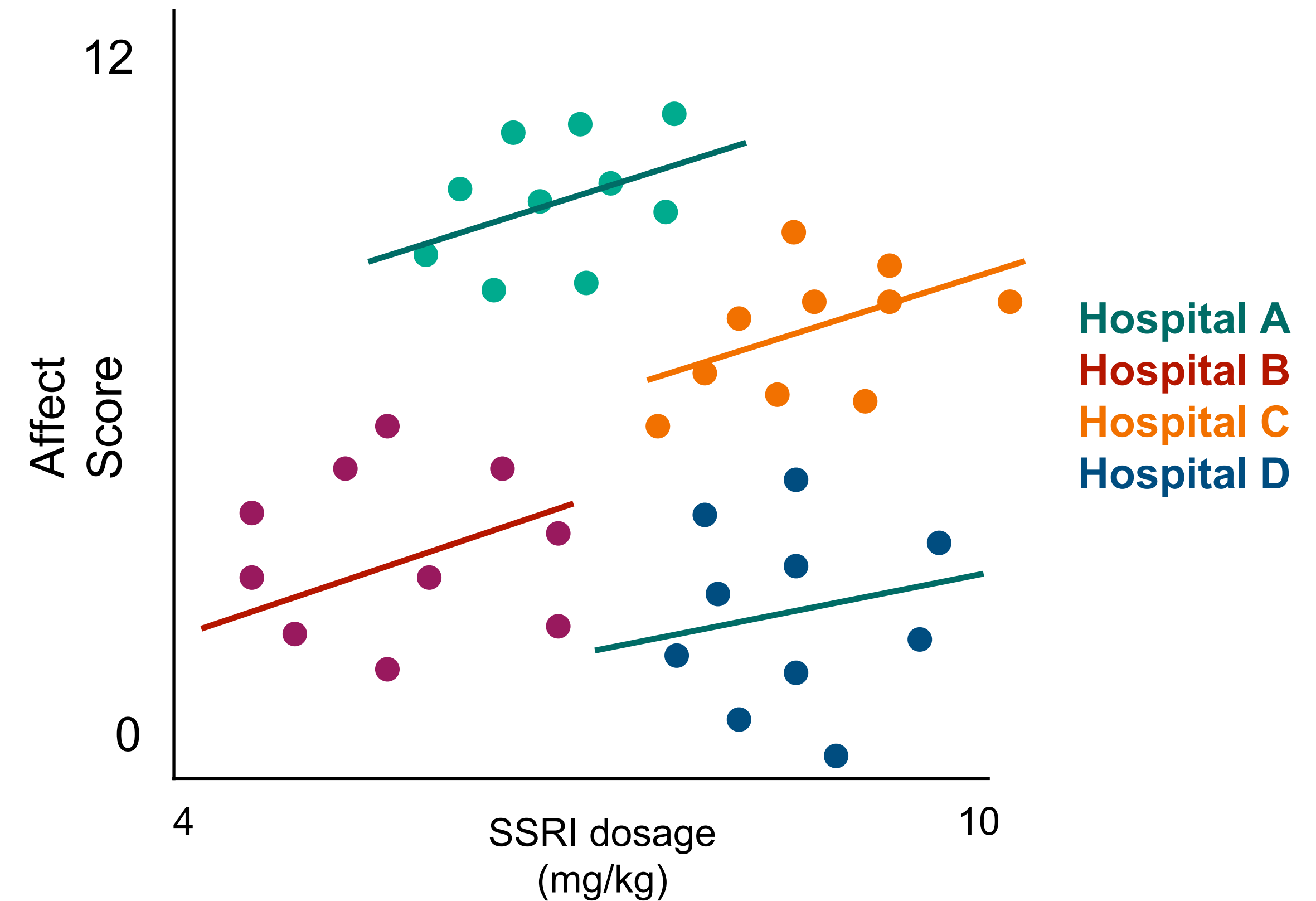
## Variables:

- $Y_{affect}$ : positive affect score
- $X_{SSRI}$ : dosage as mg/kg body weight
- $Z_{hospital}$ : hospital ID



# Types of random effects

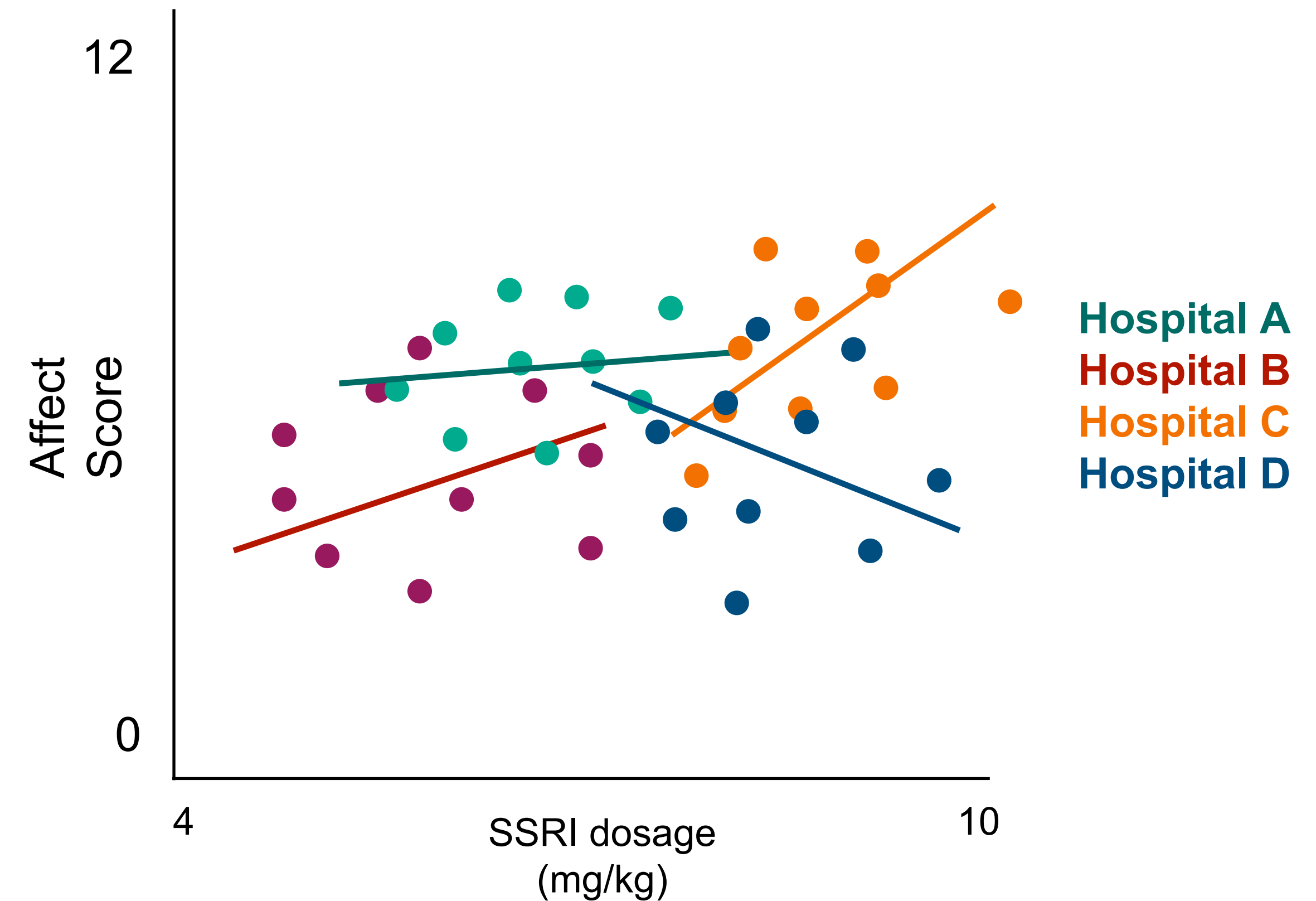
Mean effects: The random factor shifts  $\bar{Y}$   
)i.e.,  $\hat{\beta}_0$ , (but does not  
impact the nature of  $\hat{\beta}_i X$ .





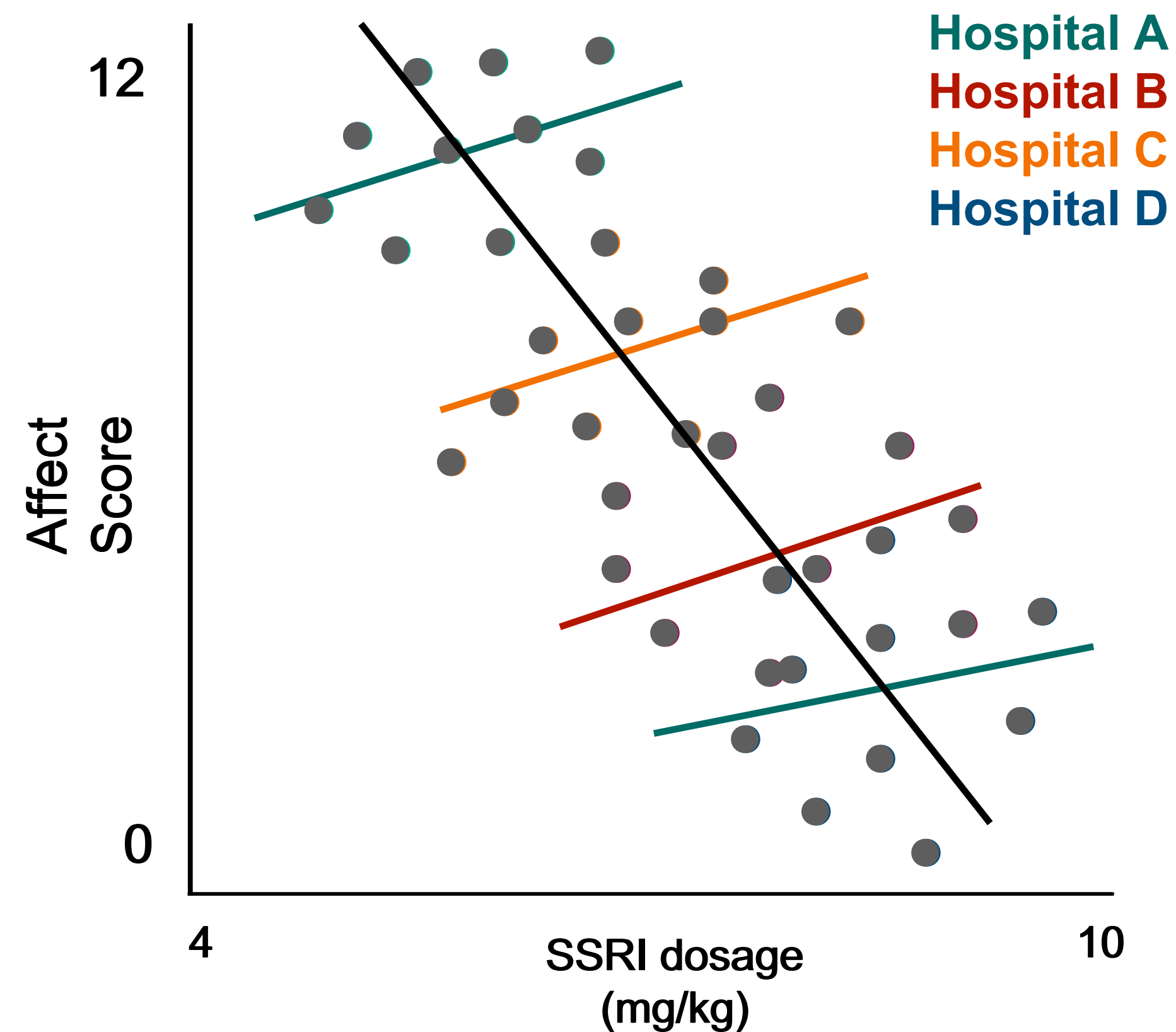
# Types of random effects

Slope effects: The random factor shifts  $\hat{\beta}_i X$ , but does not impact  $\bar{Y}$ .



# Simpson's Paradox

Across batch (group) trends conflict with within batch (group) trends



**Towards increased generalizability**



# Random variables

## Examples:

- Subject ID
- Voxel
- Neuron
- Classroom
- Data collection site
- EEG system
- Stimulus presentation software

- Image stimuli
- Testing computer
- Operating system
- Testing room

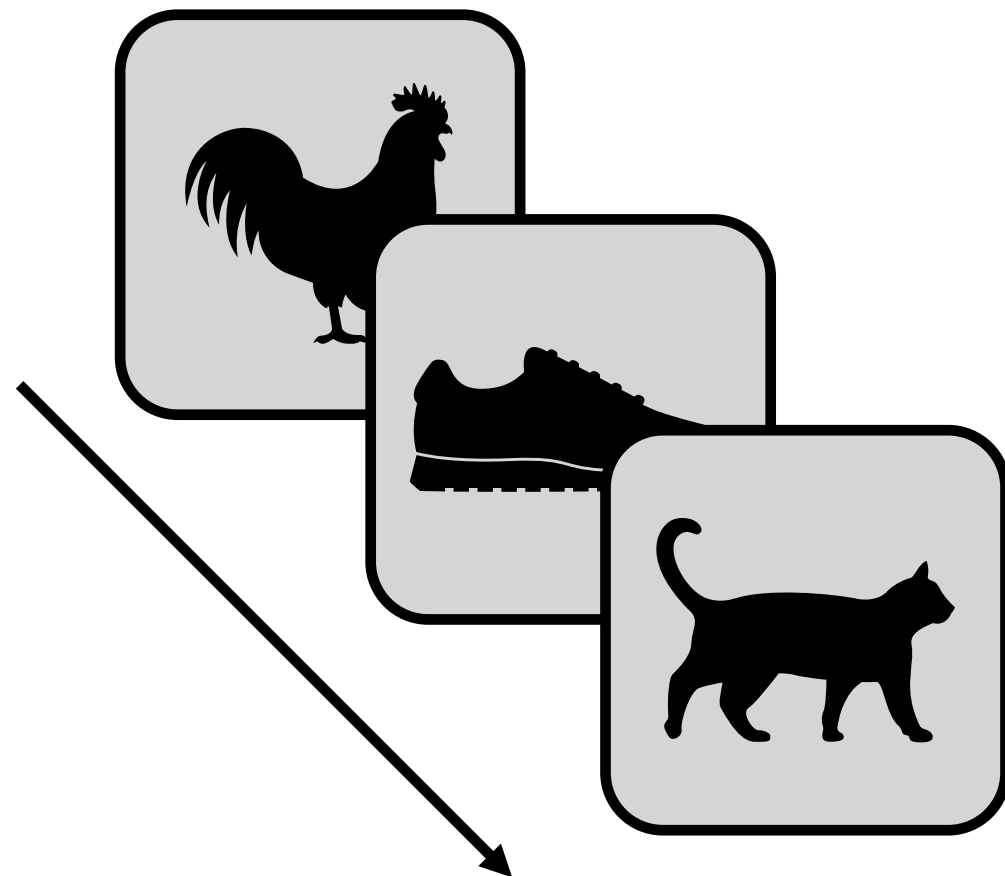
There are many random factors that contribute to experimental effects that we often overlook.

# Example: generalizable model

Q: Do people respond faster to animate objects than inanimate objects?

## Variables:

- $Y_{RT}$ : reaction time
- $X_{animacy}$ :  $\begin{cases} 1, & \text{animate} \\ 0, & \text{inanimate} \end{cases}$
- $Z_{image}$ : JPG image ID number (i.e., label each stimuli)



## Traditional model:

$$Y_{RT} = \hat{\beta}_0 + \hat{\beta}_1 X_{animacy}$$

## Full model:

$$Y_{RT} = \hat{\beta}_0 + \hat{\beta}_1 X_{animacy} + \hat{\nu}_2 Z_{image}$$

*hypotheses*      *incidental*

# Considerations

## Incidental factors:

- Image stimuli
- Testing computer
- Operating system
- Testing room
- Research assistant
- Monitor type
- Electrode
- Scanner

## Limitations:

- Statistical power (more factors = greater model variance)
- Random effects are largely uninterpretable



# Random effects vs nuisance terms

# The problem of confounds

Confounds: Variables that influence the  $Y = \hat{f}(X)$  relationship but are not needed to evaluate the central hypothesis.



Random Effects

Nuisance Factors

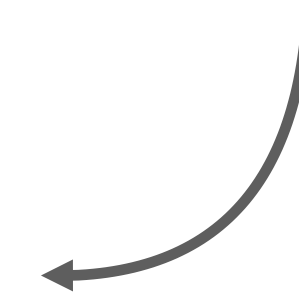


# Nuisance factors

## Two types of X:

- $X_{target}$  → factors of interest
- $X_{nuisance}$  → factors that impact  $\hat{\beta}_i X_{target}$

Units are meaningful, not random



$$Y = \hat{\beta}_0 + \hat{\beta}_1 \underbrace{X_{target}}_{hypothesis} + \hat{\beta}_2 \underbrace{X_{nuisance}}_{indidental}$$

# Collinearity

A correlation between two or more predictor variables in  $X$ .

$$Y = \hat{\beta}_0 + \hat{\beta}_1 \underbrace{X_{calories}}_{\text{hypothesis}} + \hat{\beta}_2 \underbrace{X_{height}}_{\text{indidental}}$$

$Y = \# \text{ hours of fitness}$

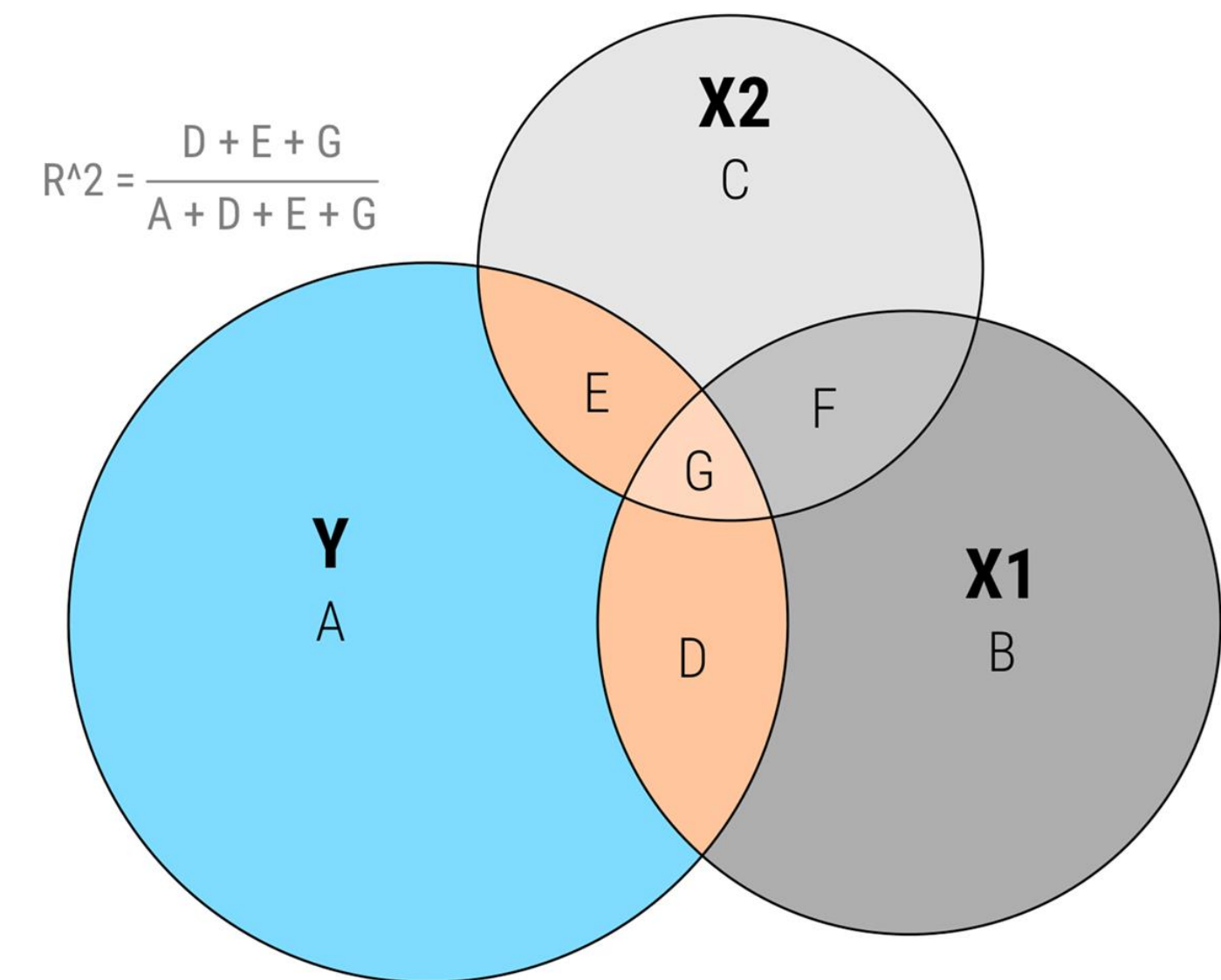
$X_1 = \text{calories}$

$X_2 = \text{height}$

Height & Calories are correlated:  $\rho(X_1, X_2) \neq 0$

Nuisance factors can leverage the effect that collinearity has on  $\hat{f}(X)$  to usefully temper confounded relationships between  $X_{target}$  and  $Y$ .

Each  $\beta_j$  reflects the contribution of  $X_j$  holding other predictors constant including the portion that overlaps with  $X_{\sim j}$ . Each  $\beta_j$  reflects the unique contribution of that variable.



# Nuisance vs. Random

OLS:  $Y = \hat{\beta}_0 + \hat{\beta}_1 X_{target} + \hat{\beta}_2 X_{nuisance}$

Objective:  $\min(\| Y - \hat{\beta}_0 - \hat{\beta}_1 X_{target} - \hat{\beta}_2 X_{nuisance} \|^2)$

Mixed Effects:  $Y = \hat{\beta}_1 X_{target} + \hat{v}_2 Z_{random}$

Objective:  $\min(\| Y - \hat{\beta}_0 - \hat{\beta}_1 X_{target} - \hat{v}_1 \Lambda_\theta Z_{random} \|^2 + \| \hat{v} \|^2)$

covariance matrix that  
explains structure in  
 $Z_{random}$



$$\hat{v} \sim N(0, \Sigma_\theta), \Sigma_\theta = \sigma \Lambda_\theta \Lambda_\theta'$$

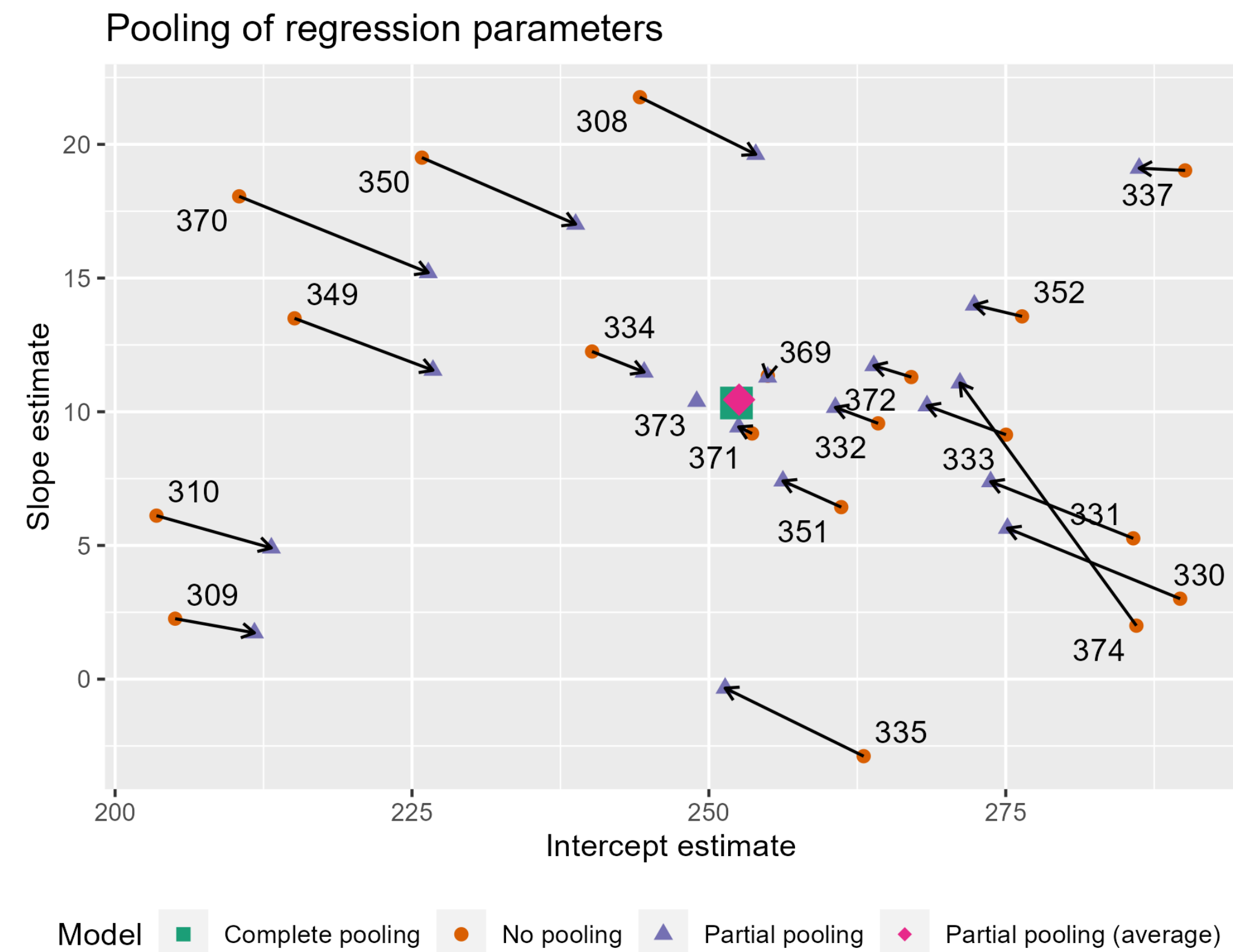
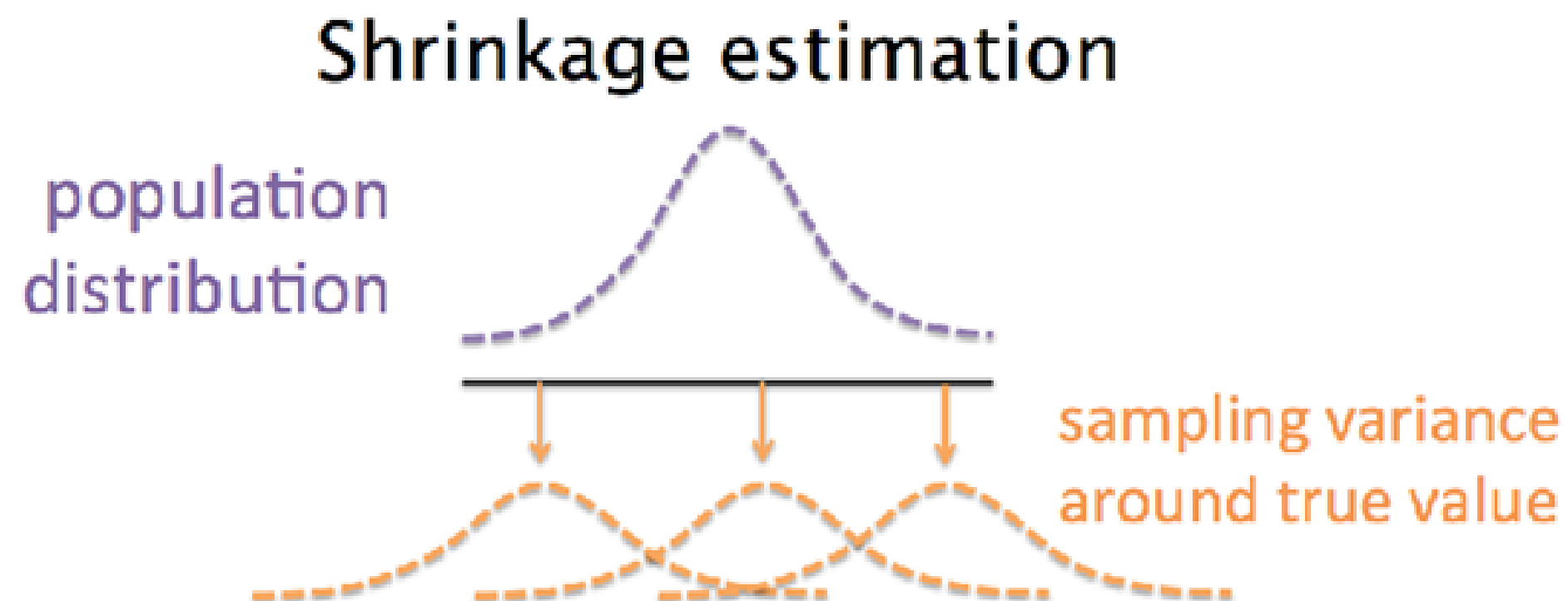


# Intuition for how Random effects are estimated

Nuisance and Random effects operate under two separate statistical assumptions

- Nuisance variables expect independence
- Random effects correct for dependence for categorical variables

Random effects ‘pull’ (through shrinkage) the mean and slope of each batch (group) towards the grand mean.



# It comes back to your question and goals

OLS:  $Y = \hat{\beta}_0 + \hat{\beta}_1 X_{\#distractors} + \hat{\beta}_2 X_{ClassRoomB}$

There is dependence among each classroom.

It's fine!

We are estimating the *difference between these **two exact rooms***. (i.e., there is no intent to generalize)

Y = performance

X1 = number of distractors

X2 = Classroom {A, B}

Mixed Effects:  $Y = \hat{\beta}_1 X_{\#distractors} + \hat{v}_2 Z_{Classroom}$

There is dependence among each classroom.

I want to generalize the effect of distractors.

Let's correct for this dependence by treating classroom as a random effect.

# Take home message

- Accounting for random effects in your model, as well as nuisance factors, can improve the generalizability of your results by accounting for incidental influences on your outcome measures.